Capstone Project Management System

A Capstone Project Presented to

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CHAPTER I
INTRODUCTION

A. PROJECT OVERVIEW

The Capstone Management System is developed to systemize the academic requirements of students, advisers, and instructors. The system manages the entire lifecycle of a capstone project, from the initial submission of titles to final archiving. This shift toward digital transformation reflects a global trend where institutions modernize administrative operations to streamline performance evaluation [1]. It is designed to replace manual document handling with a structured digital workflow for all stakeholders. By integrating automated features, the system ensures that all project milestones are recorded securely and follow established academic standards.

Aside from tracking submissions, the system facilitates formal evaluation and documentation through specialized modules. Role-based access control and encryption are critical for ensuring confidentiality and preventing unauthorized manipulation of academic data [2]. Nationally, the implementation of records tracking systems in state universities has been proven to enhance institutional transparency [3]. These platforms replace error-prone manual logbooks with automated retrieval systems to improve workflow efficiency. Ultimately, the system provides a centralized repository that maintains the integrity of the students' academic journey.

Computerized management systems have been shown to significantly reduce the workload of staff by automating report generation [4]. Locally, the development of these systems addresses specific challenges in the Davao region where stakeholders often face unorganized manual records [5]. By providing a digital management system, institutions can improve registrar

services and overall organizational performance through better safety [6]. The system also reduces administrative paperwork by automatically generating essential documents like recommendation letters and minutes. The Capstone Management System aims to make the evaluation process more efficient and accessible for the entire college.

B. PROJECT OBJECTIVES

The main goal of this project is to develop a Capstone Management System that automates the submission, routing, evaluation, and documentation of capstone project requirements by enabling role-based access for System Analyst, Adviser, Instructor, and Panel Members, ensuring efficient document management, structured evaluation, automated generation of recommendation letters, minutes, and acknowledgement receipts, and secure archiving of capstone project records.

Specifically, the system aims to develop the following:

1. User Authentication and Access Control Module
   1.1. Provide secure login and logout for authorized users.
   1.2. Display role-based dashboards based on assigned user roles.
   1.3. Restrict system access and actions based on user permissions.

2. Capstone Group and Role Management Module
   2.1. Enable the Instructor to create and manage capstone groups.
   2.2. Assign and manage student roles within each group.
   2.3. Restrict access to project records based on group membership.

3. Capstone Title Repository Module
   3.1. Allow the System Analyst to submit capstone project titles.

3.2. Enable viewing and searching of approved capstone titles.
3.3. Prevent submission of duplicate or existing project titles.
3.4. Allow the Instructor to monitor submitted and approved titles.

4. Concept Submission and Adviser Confirmation Module
   4.1. Allow the System Analyst to upload the final concept paper.
   4.2. Enable the System Analyst to select an adviser.
   4.3. Allow the Adviser to approve or decline adviser requests.
   4.4. Route the concept paper to the approved adviser.
   4.5. Automatically generate a recommendation letter upon adviser approval.

5. Defense Assignment and Panel Management Module
   5.1. Allow the Instructor to assign panel members to groups.
   5.2. Enable assignment of rooms for concept and defense presentations.
   5.3. Restrict access to submissions to assigned panel members only.
   5.4. Allow monitoring of group panel assignments and status.

6. Document Routing and Panel Evaluation Module
   6.1. Route submitted documents to assigned advisers and panel members.
   6.2. Allow panel members to input comments and evaluation feedback.
   6.3. Restrict editing or deletion of panel comments.
   6.4. Store document submissions, comments, and evaluation records.

7. Verdict Decision and Approval Module
   7.1. Allow authorized users to record the final verdict.
   7.2. Display verdict results to authorized users.
   7.3. Automatically generate an acknowledgement receipt after the final verdict.
   7.4. Store acknowledgement receipts with digital signatures.

8. Automated Minutes and Approval Sheet Module
   8.1. Automatically generate Minutes of the Meeting from panel comments.
   8.2. Allow the System Analyst to verify generated minutes.
   8.3. Enable the Adviser to approve verified minutes.
   8.4. Allow authorized users to access generated minutes and approval documents.

9. Deadline Monitoring and Submission Management Module
   9.1. Allow the Instructor to set submission deadlines.
   9.2. Monitor submission status and deadline compliance.
   9.3. Allow modification or extension of deadlines.
   9.4. Record submission activities and history.

10. Deployment Verification and Archiving Module
   10.1 Archive approved capstone documents and records.
   10.2 Allow authorized users to access archived records.
   10.3 Ensure secure storage of finalized capstone documentation.

C. SYSTEM ANALYSIS

This section describes the analytical framework used to define the operational requirements and data interactions of the Capstone Management System. It utilizes a series of visual models to illustrate how different user roles, such as the System Analyst, Adviser, Panelist, and Instructor, navigate through the project lifecycle. By mapping out the logical sequence of activities and data exchanges, the system analysis ensures that all functional modules—from user authentication to secure archiving—are seamlessly integrated. These models serve as the blueprint for developing a platform that

minimizes paperwork and automates the generation of critical academic documents. Ultimately, this analysis identifies potential bottlenecks in the manual process and provides a structured digital solution for efficient capstone management.

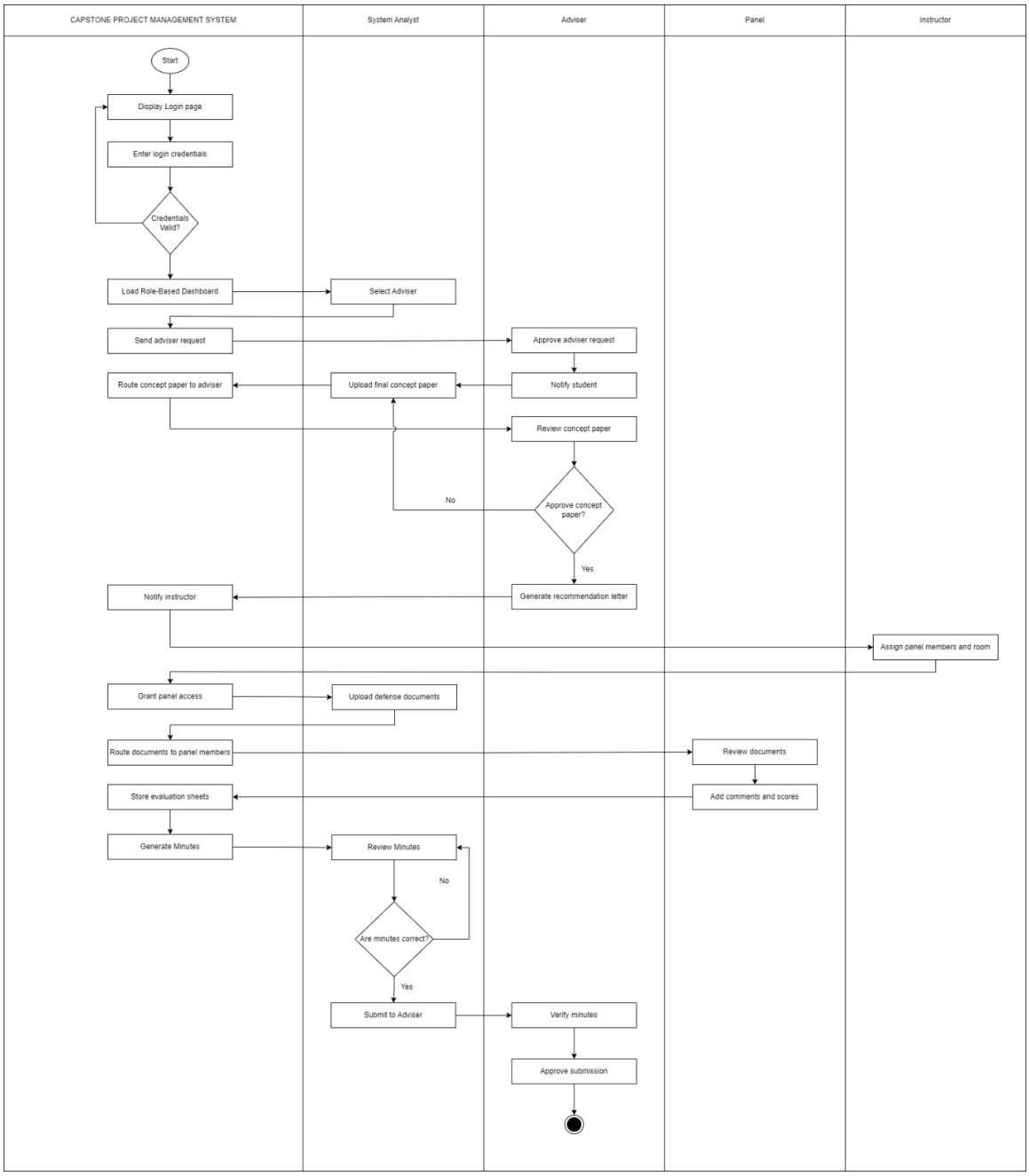

Figure 1. Concept Presentation Activity Diagram

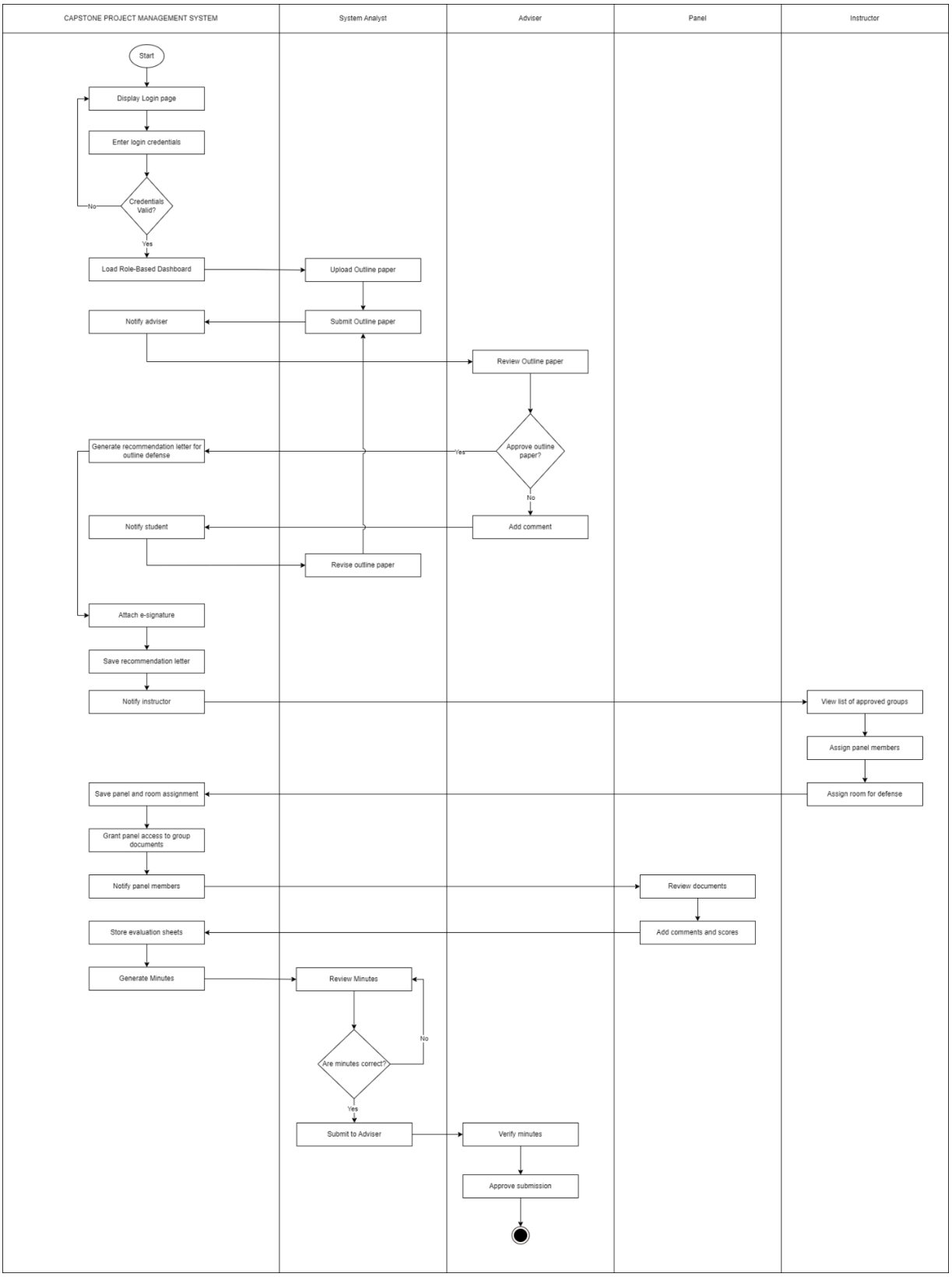

Figure SEQ Figure * ARABIC 2. Outline and Final Presentation Activity Diagram

Capstone Management System
Login
View Existing Capstone Titles
Select Adviser
Approve Adviser Selection
<<extend>>
Submit Concept Paper
Approve Concept Paper
<<extend>>
<<extend>>
Generate Recommendation Letter
Upload Documents
<<include>>
<<extend>>
View Documents
Add Comments
Review Minutes
Approve Minutes
<<extend>>
<<extend>>
Confirm Minutes
Generate Acknowledgement Receipt
<<include>>
Record Final Verdict
<<include>>
Save Evaluation Result
Set Submission Deadline
Assign Panelists
Assign Rooms
Monitor Groups
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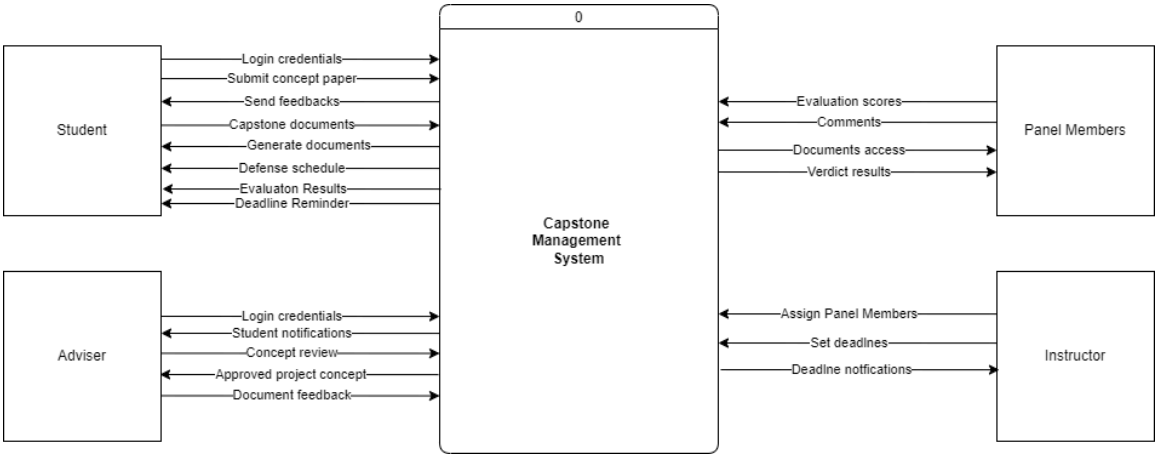
Monitor Submissions
Monitor Recommendation Letters
System Analyst
PM, Programmer, Documentarian
Adviser
Panelist
Instructor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Student
Login credentials
Submit concept paper
Send feedbacks
Capstone documents
Generate documents
Defense schedule
Evaluaton Results
Deadline Reminder
Capstone Management System
Evaluation scores
Comments
Documents access
Verdict results
Panel Members
Adviser
Login credentials
Student notifications
Concept review
Approved project concept
Document feedback
Assign Panel Members
Set deadlnes
Deadlne notfications
Instructor

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